

SECOND SEMESTER B.A./B.Sc. DEGREE EXAMINATION, APRIL 2020

(CBCSS—UG)

Mathematics

MTS 2B 02—CALCULUS OF SINGLE VARIABLE—I

(2019 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes**Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 2B 02—CALCULUS OF SINGLE VARIABLE—I

(Multiple Choice Questions for SDE Candidates)

1. Given $f(x) = 3x$ and $g(x) = x^2 - 1$. Then, the domain of $\frac{f}{g}$ is:
- (A) $[1, \infty)$. (B) $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.
(C) $(0, \infty)$. (D) $(1, \infty)$.
2. $\lim_{x \rightarrow \infty} \sin(x)$ is:
- (A) 0. (B) 1.
(C) 1 or -1. (D) Limit does not exist.
3. If f is continuous at every point of a closed interval I , then f assumes:
- (A) An absolute maximum value M but not an absolute minimum value.
(B) An absolute minimum value m but not an absolute maximum value.
(C) Both an absolute maximum value M and an absolute minimum value m .
(D) Neither an absolute maximum nor an absolute minimum.
4. On $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $f(x) = \sin(x)$ takes on:
- (A) A maximum value of 1 (once) and a minimum value of 0 (twice).
(B) A maximum value of 1 (once) and minimum value of -1.
(C) A maximum value of 1 (once) and no minimum value.
(D) A minimum value of -1 (once) and no maximum value.
5. The only domain points where a function can assume extreme values are _____.
- (A) Critical points and end points. (B) Critical points only.
(C) End points only. (D) None of the above.
6. Using which of the following reasons, can we conclude that "The Rolle's theorem cannot be applied to the function $f(x) = \tan x$ for the interval $[0, \pi]$."
- (i) There is a discontinuity at $x = \frac{\pi}{2}$ to the function $f(x) = \tan x$.
(ii) $f'(x) = \sec^2 x$ which does not exist at $x = \frac{\pi}{2}$.
- (A) Both (i) and (ii). (B) (i) only.
(C) (ii) only. (D) None of the above.

7. The value or values of c that satisfy the equation $\frac{f(b) - f(a)}{b - a} = f'(c)$ in the conclusion of Mean Value Theorem for the function $f(x) = x^2 + 2x - 1$ and the interval $[0, 1]$ is:
- (A) 1. (B) $\frac{1}{2}$.
 (C) $\frac{1}{3}$. (D) $\frac{1}{4}$.
8. At a critical point c , if f' changes from positive to negative at c ($f' > 0$ for $x < c$ and $f' < 0$ for $x > c$), then f has _____.
- (A) A local maximum value at c . (B) A local minimum value at c .
 (C) Global minimum value at c . (D) None of the above.
9. A curve is said to be concave upwards (or convex downwards) at or near P when at all points near P on it _____.
- (A) Lies *above* the tangent at P . (B) Lies *below* the tangent at P .
 (C) Lies *on* the tangent at P . (D) None of these.
10. $\lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right) =$ _____.
- (A) 0. (B) $\frac{1}{5}$.
 (C) 5. (D) ∞ .
11. $\lim_{x \rightarrow -\infty} \frac{11x + 2}{2x^3 - 1} =$ _____.
- (A) 0. (B) $\frac{11}{2}$.
 (C) ∞ . (D) $-\infty$.
12. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(4x)} :$
- (A) 1. (B) $\frac{4}{3}$.
 (C) $\frac{3}{4}$. (D) 0.
13. $\lim_{x \rightarrow 0} \frac{\sin(x)}{x^2 + 3x} :$
- (A) 1. (B) $\frac{1}{3}$.
 (C) 3. (D) $\frac{1}{4}$.

Turn over

14. Use the linear approximation of $f(x) = \sqrt{1+x}$ at $a = 0$ to estimate $\sqrt{0.95}$:
- (A) 0.942. (B) 0.995.
(C) 0.9820. (D) 0.9750.
15. If $x^2 + 2xy = y^2$, then $\frac{dy}{dx}$ is :
- (A) $\frac{x+y}{y-x}$. (B) $2x + 2y$.
(C) $\frac{x+1}{y}$. (D) $-x$.
16. If $y = 9x^2 - 4x + 3$, then $\frac{d^2y}{dx^2}$ is :
- (A) $18x - 4$. (B) -4 .
(C) 22 . (D) 18 .
17. Determine the extremas of the following function $4x^3 - 48x$:
- (A) $(2, -64)$. (B) $(2, -64)$ and $(-2, 64)$.
(C) $(2.3, -61.9)$ and $(-2, 64)$. (D) $(-2.3, 61.9)$ and $(2, -64)$.
18. The linearization of $f(x) = x^3$ at $x = 2$ is _____.
- (A) $2(6x - 7)$. (B) $2(6x + 7)$.
(C) $3x$. (D) 0 .
19. $d(\cot u) =$ _____.
- (A) $-\operatorname{cosec}^2 u \, du$. (B) $\operatorname{cosec}^2 u \, du$.
(C) $\sin u \, du$. (D) $-\sec^2 u \, du$.
20. The radius r of a circle increases from $r_0 = 10 \, m$ to $10.1 \, m$. Estimate the increase in the circle's area A by calculating dA :
- (A) $dA = 2\pi m^2$. (B) $dA = -2\pi m^2$.
(C) $dA = \pi m^2$. (D) $dA = -\pi m^2$.